LECTURE - 4

NUTRITIONAL NEEDS DURING TRAUMA (HEAD, NECK & SPINAL CORD INJURIES, etc)

HEAD INJURY

- Patients with traumatic brain injury are severely hyper metabolic and catabolic.
- The Glasgow Coma Scale (GSC)is a commonly used tool for quantifying a patients' state of consciousness.
- A score of 14-15 indicates minor head injury; 9-13 corresponds to moderate injury and less than 8 reflects severe injury.

Nutrition Therapy for HEAD INJURY

Energy

- Patients with GSC score of 4-5 often have the highest energy expenditure.
- Brain-dead patients or sedated or those who received musculoskeletal blocking agents often have lower-than predicted energy expenditure (average 14% less).
- The use of indirect calorimetry is helpful in determining the caloric requirements of these patients because overfeeding and underfeeding can be harmful.

Nutrition Therapy for HEAD INJURY

Proteins:

- Protein requirements from 1.5-2.2 g/kg body weight is estimated.

Fats:

- 30%-40% of calories for fat with omega-3 fatty acids important for brain.

Nutrition Therapy for HEAD INJURY

Vitamin, minerals and fluid:

- Urinary zinc excretion increases significantly during stress and serum zinc levels often low.
- Because salt wasting occasionally occurs in the brain-injured patient.
- Treatment may consist of restricting fluids, providing additional sodium or both.

Nutrition Therapy for HEAD INJURY

- Some brain-injured patients are unable to take oral nutrition and experience dysphagia.
- Some brain-injured patients eat rapidly and consume excessive amounts of food.

NECK INJURY

- Minor neck injuries may result from tripping, falling a short distance, or excessive twisting of the spine.
- Severe neck injuries may result from whiplash in a car accident, falls from significant heights, direct blows to the back or the top of the head, sports-related injuries, a penetrating injury such as a stab wound, or external pressure applied to the neck, such as strangulation.

NECK INJURY

- Pain from an injury may be sudden and severe.
- Bruising and swelling may develop soon after the injury.
- Acute injuries include:
  - An injury to the ligaments or muscles in the neck, such as a sprain or strain.
  - When neck pain is caused by muscle strain, you may have aches and stiffness that spread to your upper arm, shoulder, or upper back.

NECK INJURY

- Shooting pain that spreads down the arm into the hand and fingers can be a symptom of a pinched nerve (nerve root compression).
- Shooting pain is more serious if it occurs in both arms or both hands rather than just one arm or one hand.

NECK INJURY

- A fracture or dislocation of the spine.
- This can cause a spinal cord injury that may lead to permanent paralysis.
- It is important to immobilize and transport the injured person correctly to reduce the risk of permanent paralysis.
- A torn or ruptured disc - If the tear is large enough, the jellylike material inside the disc may leak out (herniate) and press against a nerve or the spinal cord (central disc herniation). patient may have a headache, feel dizzy or sick to stomach, or have pain in shoulder or down arm

Spinal cord injury (SCI)

- It includes stable fractures of the spinal column to catastrophic trans section of the spinal cord.
- Spinal cord injury is defined as a lesion in which there is no preservation of motor or sensory function more than three segments below the level of the injury.
- Health complications with this are osteoporosis, restrictive lung disease, deep vein thrombosis, joint dislocation and skin breakdown.
- Symptoms include loss of movement or feeling, numbness, tingling, difficulty controlling the muscles of the arms or legs, or loss of bowel or bladder control.

Metabolism of SCI

- Metabolism reflects the total energy used by the body and is often described in terms of calories used per day.
- Metabolism is primarily influenced by the amount of lean body mass.
- Lean body mass includes everything that isn't fat, like organs, bone, skin and muscle and is influenced by activity level.
- Metabolism typically decreases after a spinal cord injury because lean body mass is lost and patient isn’t able to be as active.
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Metabolism of SCI

- The degree of paralysis affects metabolism proportionately.
- The greater the paralysis, the fewer calories are expended.
- For more severe injuries, calories will primarily be expended for essential functions like breathing.
- For less severe injuries, activity level plays a greater role in metabolism.

Nutritional requirements for SCI

1-Carbohydrate:

- It includes fruits, vegetables and grains.
- Diet should be 55 to 60 percent carbohydrates, with emphasis on complex carbohydrates and whole grains.
- Consuming fruits and vegetables is vital because they are rich in vitamins and minerals that are necessary for fighting infection, preventing osteoporosis and avoiding pressure sores.

Nutritional requirements for SCI

1-Carbohydrate:

- Many carbohydrates are also high in fiber, which promotes healthy bowel function and discourages impaction.
- About 25 to 35 grams(g) of fiber each day is needed.
- Healthy bowel flow is crucial in spinal cord injury because clogged bowels may lead to illness, autonomic dysreflexia and death.

Nutritional requirements for SCI

2-Protein

- Healthy protein-rich foods include lean meat, fish, poultry, eggs, beans and some dairy.
- Protein contributes to immune, skin and muscle health and is important for preventing and healing skin breakdown.
- diet should be 10 to 15 percent protein, preferably from lean sources. 70g to 90g of protein per day is needed.
- **Avoid high-protein, low-carbohydrate** diets because it is a risk for kidney problems, especially since the urinary system is already more susceptible to infection after a spinal cord injury.

Nutritional requirements for SCI

3-Fat

- Fat is important for hormone production, nervous system function and vitamin absorption.
- Excessive fat intake is related to diseases like cancer, diabetes, heart disease, stroke and obesity.
- The risk for these is already higher in a spinal cord injury, so monitoring fat consumption is crucial.
- Limit fat intake to 30 percent of the total calories and choose unsaturated fats like olive oil, nuts and avocado over saturated fats like butter.
- Cooking methods like steaming and grilling can help reduce fat consumption.

Nutritional requirements for SCI

4-Weight Management

- Weight management works the same way for able bodied and paralyzed people.
- Weight gain results from taking in more calories than expended.
- Being overweight puts at greater risk for complications like pressure sores, blood clots and heart disease.
- Weight loss results from using more calories than taking in.
- Being too thin may also increase risk of pressure sores because it doesn't have enough cushion on sensitive areas.
- Weight maintenance results from consuming calories equal to the number expended.

Nutritional requirements for SCI

- Most men with SCI and women who are physically active can probably lose weight with a limit of 1,800 calories per-day.
- Most women with SCI and men with small body frames can probably lose weight with a limit of 1,500 calories per-day.
- Eat plenty of fruits and vegetables because they contain fiber, vitamins and minerals.
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Nutritional requirements for SCI

- **<u>Dietary fiber</u>** is a type of complex carbohydrate that comes from plants and can't be digested.
- Eating fiber will help maintain healthy bowel function and prevent impaction.
- It will also help manage calorie intake by having fullness for longer time.
- Vitamins and minerals are important **for preventing** osteoporosis, skin problems and disease.

Vitamins and minerals also boost immune system.

Nutritional requirements for SCI

6-Calcium

- Dairy products are the best source for calcium, which is the key nutrient in developing and maintaining bones mass.
- Calcium also helps in blood clotting and muscle and nerve functioning.
- About 1,200 milligrams (mg) of Calcium daily is needed.

Nutritional requirements for SCI

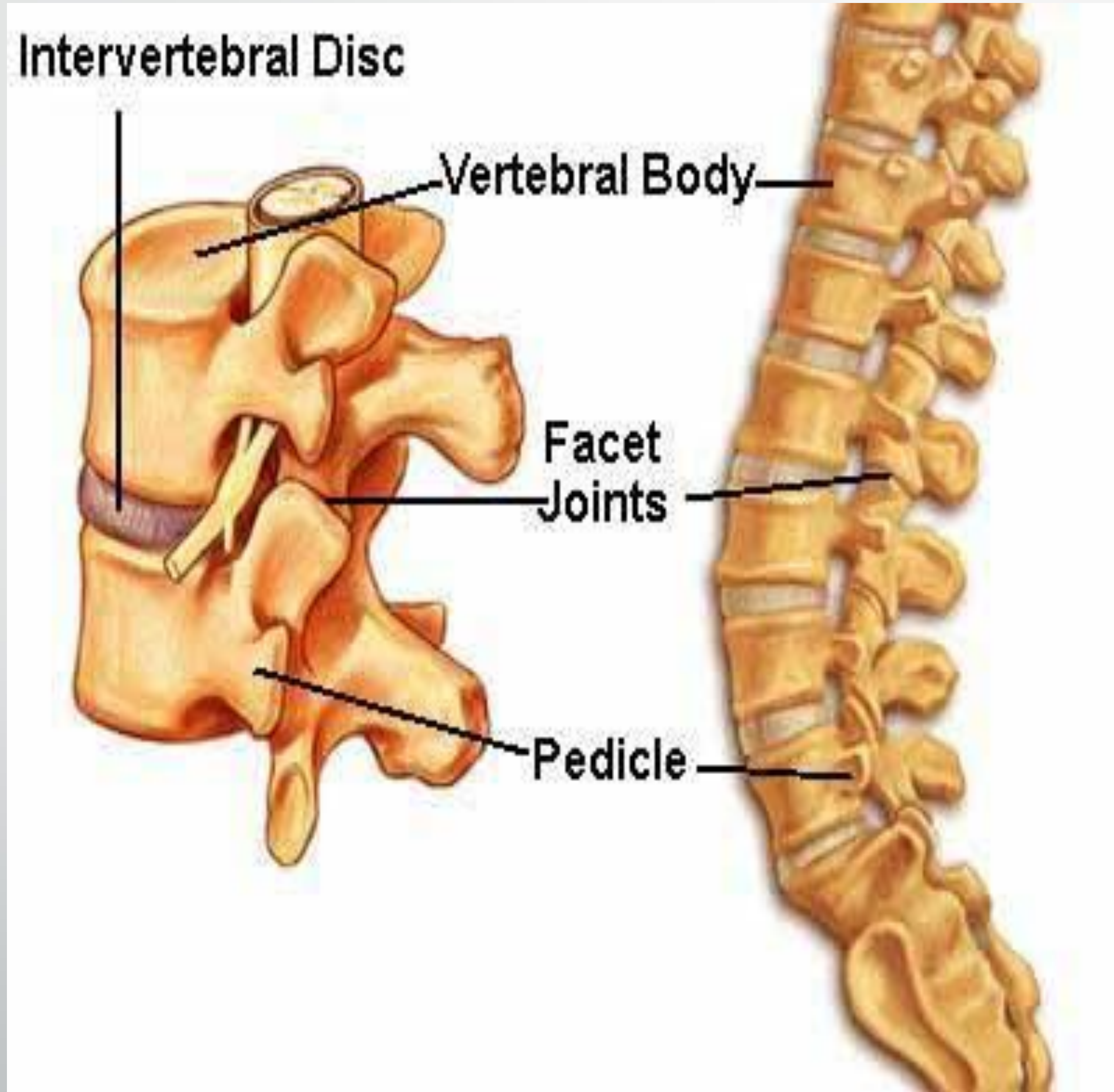
7-Sodium

- Sodium (most commonly found in salt) is needed for the regulation of body's fluid balance, contraction of muscles and conduction of nerve impulses.
- Adults should normally limit sodium to between 500mg and 1,000mg per day.
- Too much sodium causes body to retain water and puts at higher risk for swelling, heart or kidney disease and stroke.
- Limited sodium intake reduces the risk for health problems and swelling.

Nutritional requirements for SCI

8-Water:

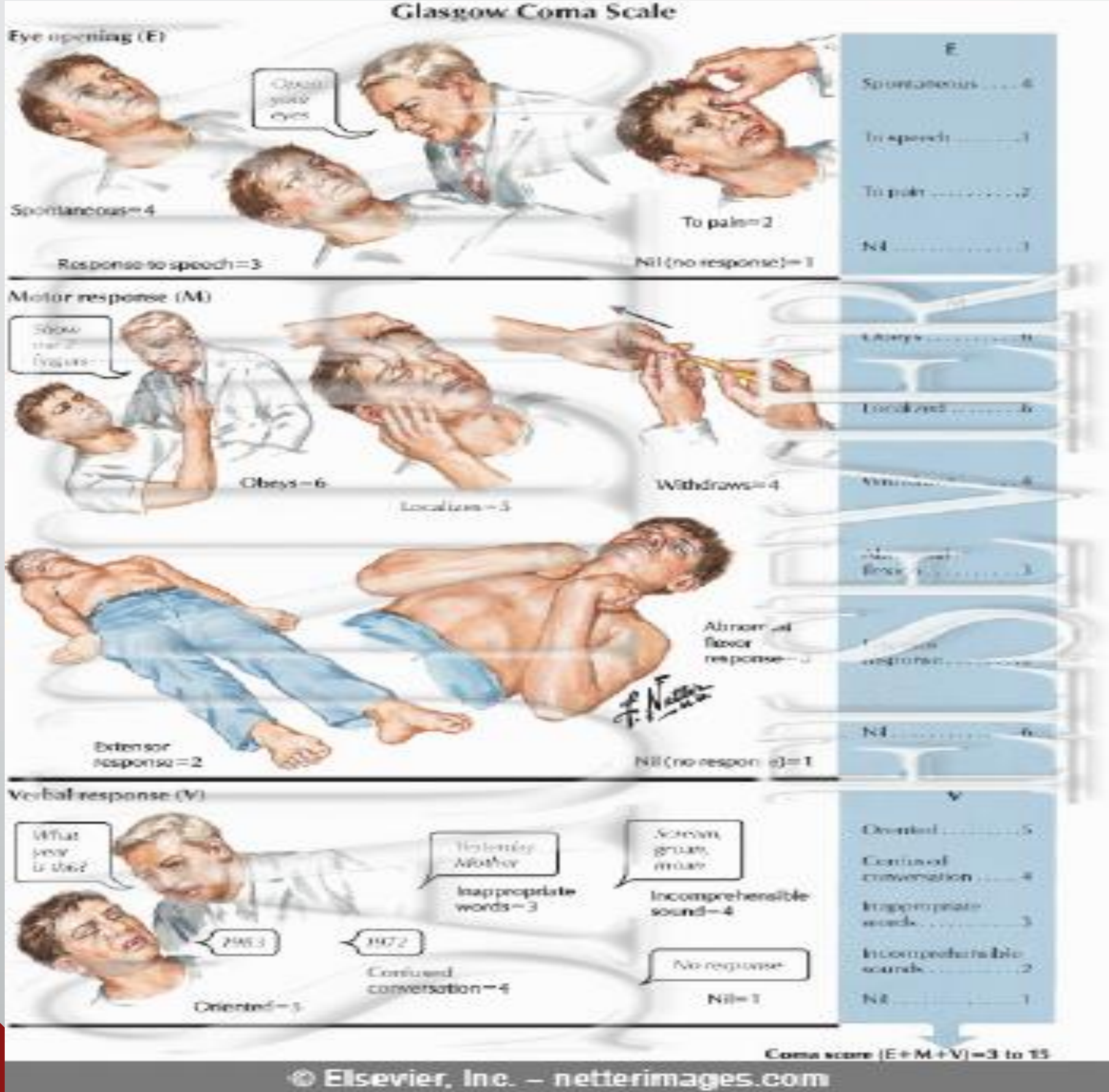
- For individuals with spinal cord injury, water helps prevent urinary tract infections as well as kidney and bladder stones.
- Water is also important in regulating bowel management.
- Although fresh vegetables and fruits are good sources for water, it is generally recommended that at least 8 cups (64 ounces) of water per day.


Intervertebral Disc
Vertebral Body
Facet
Joints
Pedicle


Spinal Cord
Nucleus Pulposus
Annulus Fibrosus
Nerve Root
Disc
Vertebra

Table 1: THE GLASGOW COMA SCALE AND SCORE

Feature	Scale Responses	Score Notation
Eye opening	Spontaneous To speech To pain None	4 3 2 1
Verbal response	Orientated Confused conversation Words (inappropriate) Sounds (incomprehensible) None	5 4 3 2 1
Best motor response	Obey commands Localise pain Flexion – Normal – Abnormal Extend None	6 5 4 3 2 1
TOTAL COMA 'SCORE'		**3/15 – 15/15**



Glasgow Coma Scale
Eye opening (E)
Spontaneous=4
Response to speech=3
To pain=2
Nil (no response)=1
Motor response (M)
Obeys=6
Withdraws=4
Extensor response=2
Verbal response (V)
Inappropriate words=3
Confused conversation=4
No response
Nil=1
Coma score (E+M+V)=3 to 15
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BOX 15-1 Medical Nutrition Therapy for Metabolically Stressed Patients

Energy requirements are highly individual and may vary widely from person to person. Total kcal requirements are dependent on the basal energy expenditure (BEE) plus the presence of trauma, surgery, infection, sepsis, and other factors. Additionally, age, height, and weight are often taken into consideration.[2]

Harris-Benedict Formula

The Harris-Benedict formula is one of the most useful and accurate for calculating basal energy requirements,[2] although it generally overestimates BEE by 5% to 15%. It is important to remember this formula uses *current (actual)* weight in the calculation.

$$\text{Wt in pounds} \div 2.2 \text{ kg} = \text{Wt in kg}$$
$$\text{Ht in inches} \times 2.54 \text{ cm} = \text{Ht in cm}$$
$$\text{Men} = 66.5 + (13.8 \times \text{Wt in kg}) + (5 \times \text{Ht in cm}) - (6.8 \times \text{Age})$$
$$\text{Women} = 65.1 + (9.6 \times \text{Wt in kg}) + (1.8 \times \text{Ht in cm}) - (4.7 \times \text{Age})$$

Once BEE has been calculated, additional kcal for activity and injury are added:

$$\text{BEE} \times \text{Activity factor (AF)} \times \text{Injury factor (IF)}$$

Protein Requirements

Additional protein is required to synthesize the proteins necessary for defense and recovery, to spare lean body mass, and to reduce the amount of endogenous protein catabolism for gluconeogenesis.

Vitamin/Mineral Needs

Needs for most vitamins and minerals increase in metabolic stress; however, no specific guidelines exist for provision of vitamins, minerals, and trace elements. It is usually believed that if the increased kcal requirements are met, adequate amounts of most vitamins and minerals are usually provided. In spite of this, vitamin C, vitamin A or beta carotene, and zinc may need special attention.

Fluid Needs

Fluid status can affect interpretation of biochemical measurements as well as anthropometry and physical examination. Fluid requirements can be estimated using several different methods.

Micronutrient Supplementation

Vitamin C: 500 to 1000 mg/daily in divided dose
Vitamin A: one multivitamin tablet containing vitamin A, one to four times daily
Zinc sulfate: 220 mg, one to three times daily

Activity	Activity Factor	Clinical Status	Energy Stress Factor	g Protein/kg Body Wt/Day
Bed rest	1.2	Elective surgery	1-1.2	1-1.5
Ambulatory	1.3	Multiple trauma	1.2-1.6	1.3-1.7
		Severe infection	1.2-1.6	
		Peritonitis	1.05-1.25	
		Multiple/long bone fractures	1.1-1.3	
		Infection with trauma	1.3-1.5	
		Sepsis	1.2-1.4	1.2-1.5
		Closed head injury	1.3	
		Cancer	1.1-1.45	
		Burns (% BSA)	1.8-2.5	
		0%-20%	1-1.5	
		20%-40%	1.5-1.85	
		40%-100%	1.85-2.05	
		Fever	1.2 per 1° C >37° C	

BSA, Body surface area.

Fluid Requirements Based On:		Water (mL)
Weight		100 mL/kg for first 10 kg
		50 mL/kg for next 10 kg
		20 mL/kg for each kg above 20 kg
Age and weight	16-30 yr (active)	40 mL/kg/day
	20-55 yr	35 mL/kg/day
	55-75 yr	30 mL/kg/day
	>75 yr	25 mL/kg/day
Energy	1 mL/kcal	
Fluid balance	Urine output + 500 mL/day	

Data from American Dietetic Association: *Manual of clinical dietetics*, ed 6, Chicago, 2000, American Dietetic Association; Helmburger DC, Ard J: *Handbook of clinical nutrition*, ed 4, St Louis, 2006, Mosby; Moore MC: *Mosby's pocket guide to nutritional care*, ed 5, St Louis, 2005, Mosby; and Winkler MF, Malone AM: Medical nutrition therapy for metabolic stress: sepsis, trauma, burns and surgery. In Mahan LK, Escott-Stump S, eds: *Krause's food, nutrition, & diet therapy*, ed 11, Philadelphia, 2004, Saunders.